



# INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

## Cloud Security (CST-109)

### Assignment 5

#### ***(Data Protection and Distributed-Cloud Security)***

#### **Public Cloud (AWS / Azure) - Provide screenshot of the below implementation**

*Flexible to use any free cloud platform (within free-tier limits)*

##### **1. Data Protection Implementation**

1. Implement *data encryption at rest* using cloud storage services (e.g., S3) and enable server-side encryption (SSE-S3). What is the encryption technique used by this?
2. Explore and implement the above using the AWS CLI.
3. What is the difference between SSE-S3 and SSE-KMS?
4. Implement *data encryption in transit* using HTTPS/TLS while accessing cloud resources (S3) and APIs.

##### **2. Data Leakage & Misconfiguration Simulation**

1. Simulate misconfigured storage access (e.g., public access to bucket) and demonstrate potential data exposure through browser and CLI.
2. Correct the configuration by blocking public access and verify secured access through browser and CLI.
3. Attempt unauthorized access from different users/roles created through **AWS IAM** and observe system behavior.
4. Check the logs in **AWS CloudTrail** to view all the failed access attempts and unauthorized user activity.

##### **3. Monitoring & Alerting**

Since you are now familiar with **AWS CloudTrail**, your task is to enable S3 data event logging and forward these logs to **AWS CloudWatch**. Using CloudWatch, analyze data access patterns, identify anomalies, and configure alerts to detect suspicious activities such as unauthorized access attempts or unusually high data downloads.

#### **Private Distributed Cloud System**

You have already successfully implemented a distributed cloud architecture. Use the architecture to implement the following.

##### **1. Data Protection Implementation**

1. Implement data encryption at rest to ensure that all data stored across multiple distributed storage nodes is securely encrypted. Each node should store only encrypted data to prevent unauthorized access.
2. Implement data encryption in transit using HTTPS/TLS to secure communication between clients, API gateway, and distributed storage nodes, ensuring data confidentiality during

transmission.

3. Design and integrate a secure key management mechanism for handling encryption and decryption keys. This may include the use of environment variables, centralized key management services, or a dedicated key management server to ensure secure key storage, access control, and rotation.

## **2. Automated Security Measures**

1. Enforce data access restrictions across storage nodes after detecting suspicious activities (e.g., repeated unauthorized access attempts), such as temporarily blocking users or limiting access to specific nodes or files.
2. Configure alerts for abnormal access behavior by monitoring access patterns across the distributed system (e.g., unusual frequency of file access, access from unknown nodes, or multiple failed authentication attempts).
3. Implement periodic key rotation and token renewal mechanisms to enhance security, ensuring that encryption keys and authentication tokens are updated regularly across all nodes without disrupting system operations.